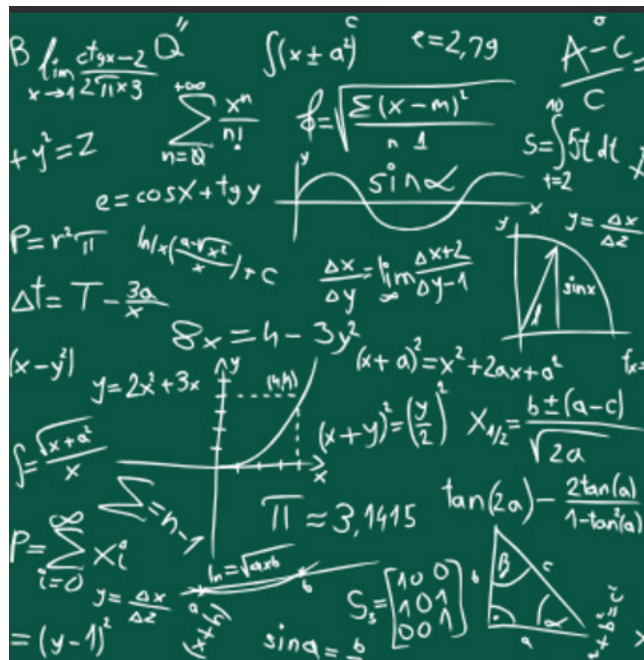


Asymptotic expansions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1	The “little o ” symbol and asymptotic expansions	2
2	Asymptotic expansions derived from the fundamental limits	5

1 The “little o ” symbol and asymptotic expansions

Definition 1: of little o

Given two functions $f(x)$ and $g(x)$, defined in a neighborhood of $c \in \mathbb{R}^*$, we say that

$$f(x) = o(g(x)) \text{ as } x \rightarrow c$$

(read “ $f(x)$ is little o of $g(x)$ ” as $x \rightarrow c$) if and only if

$$\frac{f(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

The symbol $o(g(x))$ as x tends to c does not denote a particular function $f(x)$, but any function $f(x)$ such that the ratio between $f(x)$ and $g(x)$ tends to 0 as x tends to c .

Example 1: Little o

For example:

$$x^2 = o(x) \text{ as } x \rightarrow 0 \text{ since } \frac{x^2}{x} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$x^3 = o(x) \text{ as } x \rightarrow 0 \text{ since } \frac{x^3}{x} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$x^3 = o(x^2) \text{ as } x \rightarrow 0 \text{ since } \frac{x^3}{x^2} \rightarrow 0 \text{ as } x \rightarrow 0$$

$$e^{-1/x^2} = o(x^4) \text{ as } x \rightarrow 0 \text{ since } \frac{e^{-1/x^2}}{x^4} \rightarrow 0 \text{ as } x \rightarrow 0$$

- From the definition of asymptotically equivalent functions, we have:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 1 \text{ as } x \rightarrow c$$

in these cases we have:

$$\frac{f(x)}{g(x)} - 1 \rightarrow 0 \text{ as } x \rightarrow c \quad \text{and} \quad \frac{f(x) - g(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

from the definition of little o we can then write:

$$f(x) - g(x) = o(g(x)) \text{ as } x \rightarrow c$$

We have the following *asymptotic expansion*:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = g(x) + o(g(x)) \text{ as } x \rightarrow c$$

- Moreover, we have:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c,$$

since:

$$\frac{g(x) + f(x)}{g(x)} = 1 + \frac{f(x)}{g(x)}$$

which tends to 1 as $x \rightarrow c$ if and only if $f(x)/g(x)$ tends to 0 as $x \rightarrow c$.

In such situations we say that $g(x)$ is the **principal part** of the sum $g(x) + f(x)$ and that $f(x)$ is **negligible** with respect to $g(x)$ as $x \rightarrow c$; that is:

$$f(x) = o(g(x)) \text{ as } x \rightarrow c$$

We have the following *asymptotic expansion*:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = o(g(x)) \text{ as } x \rightarrow c$$

- It follows from the definition of little o that with three functions we have:

$$f(x) = h(x) + o(g(x)) \text{ as } x \rightarrow c \iff \frac{f(x) - h(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c$$

- The “little o ” symbol behaves as follows with products:

$$f \cdot o(g) = o(f \cdot g)$$

$$o(f) \cdot o(g) = o(f \cdot g)$$

Example 2: Little o and products

For example, as $x \rightarrow c$:

$$x \cdot o(x^2) = o(x^3), \quad \frac{o(x^3)}{x} = o(x^2), \quad o(x) \cdot o(x^2) = o(x^3)$$

Some properties of the little o symbol, shown through examples:

$$o(x) \pm o(x) = o(x), \quad \text{as } x \rightarrow c$$

$$o(ax) = o(x), \quad \forall a \in \mathbb{R}, a \neq 0, \quad \text{as } x \rightarrow c$$

$$ao(x) = o(x), \quad \forall a \in \mathbb{R}, \quad \text{as } x \rightarrow c$$

$$o(x) + o(x^2) = o(x), \quad \text{as } x \rightarrow 0$$

$$o(x) + o(x^2) = o(x^2), \quad \text{as } x \rightarrow \infty$$

Definition 2: of $o(1)$

A function f is said to be infinitesimal as $x \rightarrow c$, and we write

$$f(x) = o(1) \text{ as } x \rightarrow c \text{ if } f(x) \rightarrow 0 \text{ as } x \rightarrow c$$

Example 3: $o(1)$

For example:

$$\sqrt{x} = o(1) \text{ as } x \rightarrow 0^+ \text{ since } \frac{\sqrt{x}}{1} \rightarrow 0 \text{ as } x \rightarrow 0^+$$

$$x^{-2} = o(1) \text{ as } x \rightarrow +\infty \text{ since } \frac{x^{-2}}{1} \rightarrow 0 \text{ as } x \rightarrow +\infty$$

- We have:

$$\lim_{x \rightarrow c} f(x) = \ell \iff \lim_{x \rightarrow c} (f(x) - \ell) = 0 \iff \lim_{x \rightarrow c} |f(x) - \ell| = 0$$

hence

$$\lim_{x \rightarrow c} f(x) = \ell \iff f(x) = \ell + o(1) \text{ as } x \rightarrow c$$

the last expression reads “ $f(x)$ is the sum of ℓ and of a function that is infinitesimal as $x \rightarrow c$ ”

Example 4: $o(1)$

For example:

$$x^2 = x(1 + o(1)) \text{ as } x \rightarrow 1 \text{ since } \frac{x^2}{x} = x \rightarrow 1 \text{ as } x \rightarrow 1$$

$$x^2 + 3x = x(3 + o(1)) \text{ as } x \rightarrow 0 \text{ since } \frac{x^2 + 3x}{x} = x + 3 \rightarrow 3 \text{ as } x \rightarrow 0$$

$$x^2 + 3x = x^2(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{x^2 + 3x}{x^2} = 1 + \frac{3}{x} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

$$\sqrt{x+5} = \sqrt{x}(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{\sqrt{x+5}}{\sqrt{x}} = \sqrt{1 + \frac{5}{x}} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

Some properties of the $o(1)$ symbol:

$$o(1) \pm o(1) = o(1), \text{ as } x \rightarrow c$$

$$o(1) \cdot o(1) = o(1), \text{ as } x \rightarrow c$$

$$c \cdot o(1) = o(1), \text{ as } x \rightarrow c, c \neq 0$$

$$(1 + o(1)) \cdot (1 + o(1)) = 1 + o(1), \text{ as } x \rightarrow c$$

$$\frac{1}{1 + o(1)} = 1 + o(1), \text{ as } x \rightarrow c$$

$$\frac{c}{1 + o(1)} - c = o(1), \text{ as } x \rightarrow c, c \neq 0$$

Since

$$\frac{c}{1 + o(1)} - c = c \left(\frac{1}{1 + o(1)} - 1 \right) = c(1 + o(1) - 1) = o(1)$$

Example 5: $o(1)$

We have

$$(x + o(1))^2 = x^2(1 + o(1)) \text{ as } x \rightarrow +\infty \text{ since } \frac{(x + o(1))^2}{x^2} \rightarrow 1 \text{ as } x \rightarrow +\infty$$

2 Asymptotic expansions derived from the fundamental limits

We have seen that the following fundamental limits hold:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1, \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$$

and consequently the following **asymptotic equivalences** hold as $x \rightarrow 0$:

$$\sin x \sim x, \quad 1 - \cos x \sim \frac{1}{2}x^2, \quad e^x - 1 \sim x, \quad \log(1+x) \sim x, \quad (1+x)^\alpha - 1 \sim \alpha x$$

Let us now see how to obtain the **asymptotic expansions** starting from the fundamental limits:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\sin x}{x} - 1 \rightarrow 0, \quad \frac{\sin x - x}{x} \rightarrow 0 \text{ hence } \sin x - x = o(x)$$

which gives the asymptotic expansion:

$$\sin x = x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Hence, as $x \rightarrow 0$:

$$\frac{1 - \cos x}{x^2} - \frac{1}{2} \rightarrow 0, \quad \frac{2(1 - \cos x) - x^2}{2x^2} \rightarrow 0 \text{ hence } 2(1 - \cos x) - x^2 = o(2x^2)$$

which gives the asymptotic expansion:

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\log(1+x)}{x} - 1 \rightarrow 0, \quad \frac{\log(1+x) - x}{x} \rightarrow 0 \text{ hence } \log(1+x) - x = o(x)$$

which gives the asymptotic expansion:

$$\log(1+x) = x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{e^x - 1}{x} - 1 \rightarrow 0, \quad \frac{e^x - 1 - x}{x} \rightarrow 0 \text{ hence } e^x - 1 - x = o(x)$$

which gives the asymptotic expansion:

$$e^x = 1 + x + o(x)$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R}$$

Hence, as $x \rightarrow 0$:

$$\frac{(1+x)^\alpha - 1}{x} - \alpha \rightarrow 0 \quad \frac{(1+x)^\alpha - 1 - \alpha x}{x} \rightarrow 0 \text{ hence } (1+x)^\alpha - 1 - \alpha x = o(x)$$

which gives the asymptotic expansion:

$$(1+x)^\alpha = 1 + \alpha x + o(x)$$

- These asymptotic expansions can be generalized

If $\varepsilon(x)$ is a function that tends to zero, i.e., it is an infinitesimal (it does not matter what x tends to), we have the following **asymptotic expansions** derived from the fundamental limits.

As $\varepsilon(x) \rightarrow 0$, we have:

$$\sin(\varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x)) \tag{1a}$$

$$\cos(\varepsilon(x)) = 1 - \frac{1}{2} \varepsilon^2(x) + o(\varepsilon^2(x)) \tag{1b}$$

$$\log(1 + \varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x)) \tag{1c}$$

$$e^{\varepsilon(x)} = 1 + \varepsilon(x) + o(\varepsilon(x)) \tag{1d}$$

$$(1 + \varepsilon(x))^\alpha = 1 + \alpha \varepsilon(x) + o(\varepsilon(x)) \quad \text{with } \alpha \in \mathbb{R} \tag{1e}$$